5 BAB 5 – DATA MANIPULATION LANGUAGE (DML)

5.1 IDENTITAS

Kompetensi

1. Praktikan dapat mengoperasikan perintah dml
2. Praktikan dapat memilih perintah dml yang tepat untuk memanipulasi table

Topik

1. Insert table
2. Query sederhana
3. Update table
4. Delete data
5. Query dengan kondisi

5.2 TEST AWAL

Tuliskan pada kertas jawaban pertanyaan dibawah ini. Dengan tulisan tangan, jangan lupa NIM, Nama , kelas.

1. Bagaimana cara memasukan data ke table, beri contoh !
2. Bagaimana cara menghapus data dari table, beri contoh !

5.3 INSERT TABLE

Insert merupakan perintah yang dapat digunakan untuk melakukan input data ke dalam tabel yang sudah ada.

- Perintahnya :

```
Insert Into Nama_Table Values (
            Isi_Field_1, Isi_Field_2, … , Isi_Field_N) ;
```

- Atau dengan perintah :

```
Insert Into Nama_Table (
            Nama_Field_1, Nama_Field_2, … , Nama_Field_N)
      Values
            (Isi_Field_1, Isi_Field_2, … , Isi_Field_N) ;
```

- Contoh penggunaan pada skema order entry

```
mysql> Insert Into Mahasiswa Values (,'05052652','Paijo',
                    'Jalan Wates km 11');
```

```
MariaDB [praktikum]> desc mahasiswa;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| nim      | char(11)     | YES  |     | NULL    |       |
| Nama_Mhs | char(40)     | YES  |     | NULL    |       |
| alamat   | varchar(100) | YES  |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
3 rows in set (0.01 sec)

MariaDB [praktikum]> insert into Mahasiswa values ('05052652','Painem','Jalan Wates Km 11');
Query OK, 1 row affected (0.76 sec)

MariaDB [praktikum]> select * from mahasiswa;
+----------+----------+--------------------+
| nim      | Nama_Mhs | alamat             |
+----------+----------+--------------------+
| 05052652 | Painem   | Jalan Wates Km 11 |
+----------+----------+--------------------+
1 row in set (0.00 sec)
```

Contoh dengan sintaks lain

```
MariaDB [praktikum]> insert into mahasiswa(nim,nama_mhs,alamat) values ('07080905040','Paijo','Jln Jembatan Merah no 23');
Query OK, 1 row affected (0.73 sec)

MariaDB [praktikum]> select * from mahasiswa;
+-------------+----------+--------------------------+
| nim         | Nama_Mhs | alamat                   |
+-------------+----------+--------------------------+
| 05052652    | Painem   | Jalan Wates Km 11        |
| 07080905040 | Paijo    | Jln Jembatan Merah no 23 |
+-------------+----------+--------------------------+
2 rows in set (0.00 sec)
```

Untuk memperlancar entry data , isikan table customer sehingga datanya terisi seperti dibawah ini

```
MariaDB [orderentry]> select * from customers;
+---------+---------------+--------------------+-----------+------------+----------+-------------+-------------+--------------------+
| cust_id | cust_name     | cust_address       | cust_city | cust_state | cust_zip | cust_country | cust_contact | cust_email         |
+---------+---------------+--------------------+-----------+------------+----------+-------------+-------------+--------------------+
| 10002   | Mouse House   | 333 Fromage Lane   | Columbus  | OH         | 43333    | USA          | Jerry Mouse  | NULL               |
| 10003   | Wascals       | 1 Sunny Place      | Muncie    | IN         | 42222    | USA          | Jim Jones    | rabbit@wascally.com |
| 10004   | Yosemite Place | 829 Riverside Drive | Phoenix   | AZ         | 88888    | USA          | Y Sam        | sam@yosemite.com   |
| 10005   | E Fudd        | 4545 53rd Street   | Chicago   | IL         | 54545    | USA          | E Fudd       | NULL               |
+---------+---------------+--------------------+-----------+------------+----------+-------------+-------------+--------------------+
4 rows in set (0.00 sec)
```

5.4 QUERY SEDERHANA

Select merupakan perintah yang dapat digunakan untuk :

- Menampilkan data secara keseluruhan yang terdapat di dalam table.
- Menampilkan data tertentu yang terdapat di dalam table.
- Menampilkan dan mengurutkan data secara ascending dan descending

5.4.1 Menampilkan Data Secara Keseluruhan

Jika ingin menampilkan data data secara keseluruhan yang terdapat di dalam table, misalnya table Mhs_2 secara keseluruhan.

Perintahnya :

```
Select * From Nama_Table;
```

Contoh :

```
mysql> Select * From Mahasiswa;
```

```
MariaDB [praktikum]> select * from mahasiswa;
+-------------+----------+--------------------------+
| nim         | Nama_Mhs | alamat                   |
+-------------+----------+--------------------------+
| 05052652    | Painem   | Jalan Wates Km 11        |
| 07080905040 | Paijo    | Jln Jembatan Merah no 23 |
+-------------+----------+--------------------------+
2 rows in set (0.00 sec)
```

Menampilkan Kolom Data Tertentu

Jika hanya ingin menampilkan beberapa field tertentu dalam suatu table. Misalkan dari data yang terdapat pada tabel Mahasiswa yang mempunyai Field (NIM, Nama_Mhs, Alamat) dan hanya akan menapilkan NIM dan Nama_Mhs.

Perintahnya :

```
Select Nama_Field_1, … , Nama_Field_N From Nama_Table;
```

Contoh :

```
mysql> Select NIM, Nama_Mhs From Mahasiswa;
```

```
MariaDB [praktikum]> select nim,nama_mhs from mahasiswa;
+-------------+----------+
| nim         | nama_mhs |
+-------------+----------+
| 05052652    | Painem   |
| 07080905040 | Paijo    |
+-------------+----------+
2 rows in set (0.14 sec)
```

Menampilkan Baris Data Tertentu

Jika hanya ingin menampilkan beberapa baris tertentu dalam suatu table. Misalkan dari data yang terdapat pada tabel Mahasiswa ingin menampilkan baris tertentu maka akan ditambahkan kondisi pada clause setelah where.

Perintahnya :

```
Select * from Nama_Table where Kondisi;
```

Contoh :

```
mysql> Select NIM, Nama_Mhs From Mahasiwa Where Nim=05052652;
```

```
MariaDB [praktikum]> select * from mahasiswa where nim=05052652;
+----------+----------+------------------+
| nim      | Nama_Mhs | alamat           |
+----------+----------+------------------+
| 05052652 | Painem   | Jalan Wates Km 11 |
+----------+----------+------------------+
1 row in set (3.23 sec)
```

5.5 UPDATE TABLE

Update merupakan perintah yang dapat digunakan untuk melakukan perubahan terhadap data yang sudah ada/dibuat. Latihan setelah sub bab ini kita gunakan **skema order entry**

- Perintahnya :

```
Update Nama_Table Set Nama_Field = 'Data_Baru'
                  Where Nama_Field_Key = 'Data_Key';
```

- Contoh :

```
mysql> Update customers Set cust_address = 'Gejayan Yogya',cust_country=' INA'
                  Where cust_id='10002';
```

```
MariaDB [orderentry]> update customers set cust_address='Gejayan Yogya',cust_country='INA' where cust_id='10002';
Query OK, 1 row affected (0.42 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [orderentry]> select * from customers;
+---------+---------------+--------------------+-----------+------------+---------+-------------+-------------+--------------------+
| cust_id | cust_name     | cust_address       | cust_city | cust_state | cust_zip | cust_country | cust_contact | cust_email         |
+---------+---------------+--------------------+-----------+------------+---------+-------------+-------------+--------------------+
| 10002   | Mouse House   | Gejayan Yogya      | Columbus  | OH         | 43333   | INA          | Jerry Mouse | NULL               |
| 10003   | Wascals       | 1 Sunny Place      | Muncie    | IN         | 42222   | USA          | Jim Jones   | rabbit@wascally.com |
| 10004   | Yosemite Place | 829 Riverside Drive | Phoenix   | AZ         | 88888   | USA          | Y Sam       | sam@yosemite.com   |
| 10005   | E Fudd        | 4545 53rd Street   | Chicago   | IL         | 54545   | USA          | E Fudd      | NULL               |
+---------+---------------+--------------------+-----------+------------+---------+-------------+-------------+--------------------+
4 rows in set (0.00 sec)
```

5.6 DELETE DATA

Delete merupakan perintah yang dapat digunakan untuk menghapus data yang terdapat di dalam tabel.

- Perintahnya :

```
Delete From Nama_Table Where Nama_Field_Key;
```

- Contoh :

```
mysql> Delete From Mhs_2 Where ID='5';
```

```
MariaDB [orderentry]> select * from customers;
+---------+---------------+---------------------+-----------+-----------+----------+--------------+--------------+---------------------+
| cust_id | cust_name     | cust_address        | cust_city | cust_state | cust_zip | cust_country | cust_contact | cust_email          |
+---------+---------------+---------------------+-----------+-----------+----------+--------------+--------------+---------------------+
| 10002   | Mouse House   | Gejayan Yogya       | Columbus  | OH         | 43333    | INA          | Jerry Mouse  | NULL                |
| 10003   | Wascals       | 1 Sunny Place       | Muncie    | IN         | 42222    | USA          | Jim Jones    | rabbit@wascally.com |
| 10004   | Yosemite Place | 829 Riverside Drive | Phoenix   | AZ         | 88888    | USA          | Y Sam        | sam@yosemite.com    |
| 10005   | E Fudd        | 4545 53rd Street    | Chicago   | IL         | 54545    | USA          | E Fudd       | NULL                |
| 10006   | Ponidi        | Jln Wates km 10     | Yogyakarta |           |          |              |              |                     |
+---------+---------------+---------------------+-----------+-----------+----------+--------------+--------------+---------------------+
5 rows in set (0.00 sec)

MariaDB [orderentry]> delete from customers where cust_id='10006';
Query OK, 1 row affected (0.24 sec)

MariaDB [orderentry]> select * from customers;
+---------+---------------+---------------------+-----------+-----------+----------+--------------+--------------+---------------------+
| cust_id | cust_name     | cust_address        | cust_city | cust_state | cust_zip | cust_country | cust_contact | cust_email          |
+---------+---------------+---------------------+-----------+-----------+----------+--------------+--------------+---------------------+
| 10002   | Mouse House   | Gejayan Yogya       | Columbus  | OH         | 43333    | INA          | Jerry Mouse  | NULL                |
| 10003   | Wascals       | 1 Sunny Place       | Muncie    | IN         | 42222    | USA          | Jim Jones    | rabbit@wascally.com |
| 10004   | Yosemite Place | 829 Riverside Drive | Phoenix   | AZ         | 88888    | USA          | Y Sam        | sam@yosemite.com    |
| 10005   | E Fudd        | 4545 53rd Street    | Chicago   | IL         | 54545    | USA          | E Fudd       | NULL                |
+---------+---------------+---------------------+-----------+-----------+----------+--------------+--------------+---------------------+
4 rows in set (0.00 sec)
```

5.7 QUERY DENGAN KONDISI

Query dengan perbandingan kondisi bentuk umumnya adalah sbb :

Select * from Nama_Table where **Kondisi**;

Pada bagian kondisi bisa diberikan berbagai value misalnya salah satunya dengan beberapa operator relasional

Operator Relasional

Operator relasional merupakan operator yang digunakan untuk membandingkan antara dua buah nilai dalam suatu table.

Operator	Keterangan
=	Sama dengan
>	Lebih besar dari
<	Lebih kecil dari
>=	Lebih besar dari sama dengan
<=	Lebih kecil dari sama dengan
<>	Lebih kurang

Perintahnya :

Select * **From** Nama_Table
Where Nama_Field [Operator Relasional] Ketentuan;

contoh :

select * from customer where cust_id = ‘10003’ or cust_name =’ wascals;

```
MariaDB [orderentry]> select * from customers where cust_id='10003' or cust_name='Wascals';
+---------+-----------+--------------+-----------+------------+----------+--------------+--------------+--------------------+
| cust_id | cust_name | cust_address | cust_city | cust_state | cust_zip | cust_country | cust_contact | cust_email         |
+---------+-----------+--------------+-----------+------------+----------+--------------+--------------+--------------------+
| 10003   | Wascals   | 1 Sunny Place | Muncie    | IN         | 42222    | USA          | Jim Jones    | rabbit@wascally.com |
+---------+-----------+--------------+-----------+------------+----------+--------------+--------------+--------------------+
1 row in set (0.01 sec)
```

artinya menampilkan data customer yang mempunya id 10003 atau namanya wascals.

```
select prod_name from products where prod_price >=20 ;
```

artinya menampilkan data barang barang yang harganya lebih dari 20.

```
MariaDB [orderentry]> select prod_name from products where prod_price>=20;
+--------------+
| prod_name    |
+--------------+
| JetPack 1000 |
| JetPack 2000 |
| Safe         |
+--------------+
3 rows in set (0.06 sec)
```

Bisa dicoba untuk query dengan berbagai kondisi yang lain. Silahkan tanya asisten untuk lebih lanjut.

5.8 TEST AKHIR

1. Berdasarkan Database order entry dan tabel yang telah dibuat sebelumnya, Isikan tabel berikut ini sesuai dengan isian sbb

    a. Tabel customers

```
MariaDB [orderentry]> select * from customers;
+---------+---------------+--------------------+-----------+------------+----------+--------------+--------------+--------------------+
| cust_id | cust_name     | cust_address       | cust_city | cust_state | cust_zip | cust_country | cust_contact | cust_email         |
+---------+---------------+--------------------+-----------+------------+----------+--------------+--------------+--------------------+
| 10002   | Mouse House   | 333 Fromage Lane   | Columbus  | OH         | 43333    | USA          | Jerry Mouse  | NULL               |
| 10003   | Wascals       | 1 Sunny Place      | Muncie    | IN         | 42222    | USA          | Jim Jones    | rabbit@wascally.com |
| 10004   | Yosemite Place | 829 Riverside Drive | Phoenix   | AZ         | 88888    | USA          | Y Sam        | sam@yosemite.com   |
| 10005   | E Fudd        | 4545 53rd Street   | Chicago   | IL         | 54545    | USA          | E Fudd       | NULL               |
+---------+---------------+--------------------+-----------+------------+----------+--------------+--------------+--------------------+
4 rows in set (0.00 sec)
```

    b. Tabel Products

```
MariaDB [orderentry]> select * from products;
+---------+---------+----------------+------------+-----------------------------------------------------------------+
| prod_id | vend_id | prod_name      | prod_price | prod_desc                                                       |
+---------+---------+----------------+------------+-----------------------------------------------------------------+
| ANV01   | 1001    | .5 ton anvil   |          6 | .5 ton anvil, black, complete with handy hook                   |
| ANV02   | 1001    | 1 ton anvil    |         10 | 1 ton anvil, black, complete with handy hook and carrying case  |
| ANV03   | 1001    | 2 ton anvil    |         15 | 2 ton anvil, black, complete with handy hook and carrying case  |
| DTNTR   | 1003    | Detonator      |         13 | Detonator (plunger powered), fuses not included                 |
| FB      | 1003    | Bird seed      |         10 | Large bag (suitable for road runners)                           |
| FC      | 1003    | Carrots        |          3 | Carrots (rabbit hunting season only)                            |
| FU1     | 1002    | Fuses          |          3 | 1 dozen, extra long                                             |
| JP1000  | 1005    | JetPack 1000   |         35 | JetPack 1000, intended for single use                           |
| JP2000  | 1005    | JetPack 2000   |         55 | JetPack 2000, multi-use                                         |
| OL1     | 1002    | Oil can        |          9 | Oil can, red                                                    |
| SAFE    | 1003    | Safe           |         50 | Safe with combination lock                                      |
| SLING   | 1003    | Sling          |          4 | Sling, one size fits all                                        |
| TNT1    | 1003    | TNT (1 stick)  |          3 | TNT, red, single stick                                          |
| TNT2    | 1003    | TNT (5 sticks) |         10 | TNT, red, pack of 10 sticks                                     |
+---------+---------+----------------+------------+-----------------------------------------------------------------+
14 rows in set (0.00 sec)
```

    c. Tabel vendors

```
MariaDB [orderentry]> select * from vendors;
+---------+---------------+----------------+------------+-----------+----------+--------------+
| vend_id | vend_name     | vend_address   | vend_city  | vend_state | vend_zip | vend_country |
+---------+---------------+----------------+------------+-----------+----------+--------------+
| 1001    | Anvils R Us   | 123 Main Street | Southfield | MI        | 48075    | USA          |
| 1002    | LT Supplies   | 500 Park Street | Anytown    | OH        | 44333    | USA          |
| 1003    | ACME          | 555 High Street | Los Angeles | CA       | 90046    | USA          |
| 1004    | Furball Inc.  | 1000 5th Avenue | New York   | NY        | 11111    | USA          |
| 1005    | Jet Set       | 42 Galaxy Road  | London     | NULL      | N16 6PS  | England      |
| 1006    | Jouets Et Ours | 1 Rue Amusement | Paris     | NULL      | 45678    | France       |
+---------+---------------+----------------+------------+-----------+----------+--------------+
6 rows in set (0.00 sec)
```

d. Tabel orderitems

```
MariaDB [orderentry]> select * from orderitems;
+-----------+------------+---------+----------+
| order_num | order_item | prod_id | quantity |
+-----------+------------+---------+----------+
|     20005 |          1 | ANV01   |       10 |
|     20005 |          2 | ANV02   |        3 |
|     20005 |          3 | TNT2    |        5 |
|     20005 |          4 | FB      |        1 |
|     20006 |          1 | JP2000  |        1 |
|     20007 |          1 | TNT2    |      100 |
|     20008 |          1 | FC      |       50 |
|     20009 |          1 | FB      |        1 |
|     20009 |          2 | OL1     |        1 |
|     20009 |          3 | SLING   |        1 |
|     20009 |          4 | ANV03   |        1 |
+-----------+------------+---------+----------+
11 rows in set (0.00 sec)
```

2. Setelah keempat table diisi tampilkan query berikut ini
    a. Ubahlah data salah satu customer dengan perintah update
    b. Hapuslah data salah satu vendor atau product dengan perintah delete
    c. Tampilkan nomer order dan jumlah nya
    d. Tampilkan data customer yang berasal dari ‘USA’.
3. Buatlah laporan praktikum dengan ketentuan sebagai berikut :
    a. Nama file laporan : PrakDB_Bab5-NIM.odt
    b. Isi file laporan :
        i. Source SQL
        ii. Screenshot CMD
    c. Simpan di directory “PrakDB-NIM” .
4. Jawaban